

An AI-Based Dengue Outbreak Prediction and Intervention Optimization System for the Colombo District, Sri Lanka

Shashika Lokuliyana
Department of Computer System
Engineering
Sri Lanka Institute of Information
Technology
Malabe , Sri Lanka
shashika.l@slit.lk

Narmada Gamage
Department of Computer System
Engineering
Sri Lanka Institute of Information
Technology
Malabe , Sri Lanka
narmada.g@slit.lk

Wickramarachchi T.T
Department of Information
Technology
Sri Lanka Institute of Information
Technology
Malabe , Sri Lanka
it22243980@my.slit.lk

Hettiarachchi A.T
Department of Information Technology
Sri Lanka Institute of Information
Technology
Malabe , Sri Lanka
it22165848@my.slit.lk

Keenavinna K.A.I.U
Department of Information Technology
Sri Lanka Institute of Information
Technology
Malabe , Sri Lanka
it22214034@my.slit.lk

Adiithya A.A.H
Department of Information Technology
Sri Lanka Institute of Information
Technology
Malabe , Sri Lanka
it22281982@my.slit.lk

Abstract— Dengue remains a significant public health concern in Sri Lanka, where timely identification of outbreak risks and appropriate intervention planning are essential for effective disease control. This research presents an integrated Artificial Intelligence (AI)-based decision-support system for dengue outbreak prediction and intervention optimization in the Colombo District. The proposed system combines Machine Learning (ML), Explainable Artificial Intelligence (XAI), Reinforcement Learning (RL), and Natural Language Processing (NLP) within a unified mobile application. A Random Forest classifier is used to predict dengue outbreak risk as Low, Medium, or High using temporal and meteorological variables, achieving an overall prediction accuracy of 86.35%. SHapley Additive exPlanations (SHAP) are incorporated to improve the transparency of the prediction model by providing local and global explanations of the factors influencing individual and overall predictions. A Q-Learning agent recommends context-dependent dengue interventions based on dengue cases, rainfall, and temperature conditions. In addition, a DistilBERT-based NLP component supports multilingual dengue-related information classification and misinformation detection. The system components are integrated through backend services and a Flutter mobile application that provides outbreak predictions, probability visualizations, XAI interpretations, intervention recommendations, temporal and geographic monitoring, chatbot assistance, and automated alerts. The proposed system demonstrates the feasibility of integrating predictive, explanatory, prescriptive, and communicative AI capabilities within a unified decision-support platform to support more proactive and data-driven dengue surveillance and control.

Keywords— *Dengue outbreak prediction, Machine Learning, Random Forest, Explainable Artificial Intelligence, SHAP, Reinforcement Learning, Q-Learning, Natural Language Processing, DistilBERT, Decision Support System.*

I. INTRODUCTION

Dengue remains a major public health concern in tropical and subtropical regions, including Sri Lanka. The disease places considerable pressure on healthcare systems, particularly during periods of increased transmission and outbreaks [1]. Effective dengue management therefore

requires timely surveillance, early identification of potential outbreak conditions, and appropriate preventive interventions.

Machine Learning has increasingly been investigated as a method for supporting dengue prediction and surveillance. Previous studies have demonstrated that historical dengue patterns, temporal variables, and environmental conditions can be analyzed to estimate potential future outbreak risks [2], [3]. Environmental factors such as rainfall, temperature, and humidity can influence mosquito breeding and dengue transmission patterns, making meteorological information valuable for predictive modelling. Among the Machine Learning techniques used for classification and prediction, ensemble learning methods have received considerable attention. Random Forest combines multiple decision trees to improve predictive performance and reduce the influence of individual trees [4]. XGBoost is another widely used tree-based method that has demonstrated strong performance in structured data analysis and predictive modelling [5]. However, although such models can generate useful predictions, the complexity of Machine Learning models may make it difficult for users to understand the factors influencing the generated outcomes.

Explainable Artificial Intelligence has therefore emerged as an important approach for improving the transparency and interpretability of complex prediction models. SHapley Additive exPlanations provide a theoretically grounded method for explaining model predictions by assigning contribution values to individual input features [6]. SHAP can provide both local and global explanations, allowing users to understand the factors influencing a specific prediction as well as the overall importance of variables across the model.

However, predicting an outbreak and explaining its causes do not directly determine the most appropriate intervention. Reinforcement Learning supports sequential

decision-making by enabling an agent to learn suitable actions through environmental interaction and reward-based feedback [7], [15]. In dengue management, this approach can support the selection of context-dependent preventive and control measures. Natural Language Processing can further improve access to dengue-related health information. Transformer-based architecture provides strong contextual language understanding [8], [16], while DistilBERT offers a more computationally efficient alternative while retaining substantial language capabilities [9].

Despite the potential of these technologies, many existing studies investigate prediction, explainability, intervention recommendation, and health-information processing as separate functions. Therefore, an important research gap exists in integrating these capabilities into a unified dengue decision-support platform. The proposed system addresses this gap by combining dengue outbreak prediction, SHAP-based explanation, Reinforcement Learning-based intervention recommendation, visualization, and NLP-based health intelligence within a single mobile application.

II. LITERATURE REVIEW

A. Machine Learning-Based Dengue Outbreak Prediction

Machine Learning has been widely investigated for dengue prediction and epidemiological modelling. Hoyos *et al.* identified prediction and epidemiological modelling as important Machine Learning applications in dengue research [2]. Previous studies have also demonstrated the usefulness of historical dengue patterns and environmental variables for forecasting [3]. Tree-based models such as Random Forest and XGBoost are particularly suitable for structured temporal and environmental datasets. Random Forest can provide strong predictive performance while reducing the effects of noise and overfitting [4], [17], whereas XGBoost uses gradient boosting to improve prediction performance [5]. These studies demonstrate the potential of data-driven approaches for early dengue risk identification. However, accurate predictions alone are insufficient when decision-makers also need to understand the factors influencing the predicted outcome.

B. Explainable Artificial Intelligence for Dengue Prediction

Complex Machine Learning models may produce accurate predictions while providing limited information about the reasoning behind the results. This limitation is particularly important in public-health applications, where users may require understandable evidence before taking action. SHAP provides a unified framework for interpreting Machine Learning predictions by assigning contribution values to individual features [6]. Local explanations can identify the variables that increase or decrease the likelihood of a specific predicted outcome, while global explanations can identify the overall importance of variables across the model. Therefore, SHAP provides a suitable method for improving the transparency of dengue outbreak predictions. However, explanation alone does not recommend the most

appropriate intervention following the identification of a potential outbreak risk.

C. Reinforcement Learning for Intervention Recommendation

Reinforcement Learning is a computational approach in which an agent learns an appropriate decision policy by interacting with an environment and receiving rewards for its actions [7], [15]. Q-Learning is a widely used Reinforcement Learning algorithm that estimates the expected value of performing actions under specific states. This approach can support context-dependent intervention selection because different combinations of dengue cases and environmental conditions may require different public-health responses. Rather than relying on a single fixed recommendation, a Q-Learning agent can learn a decision policy that associates different states with appropriate actions. In the proposed system, Reinforcement Learning extends the functionality of dengue prediction from identifying potential risk to recommending suitable preventive or control actions.

D. Dengue Health Intelligence and Misinformation Analysis

Access to reliable dengue-related health information is essential for disease prevention and public health decision-making. However, conventional health-information systems may provide limited support for multilingual users, follow-up questions, and the identification of misleading health claims. Transformer-based language models have improved contextual text understanding, with BERT introducing deep bidirectional language representation [8] and DistilBERT providing a more computationally efficient alternative [9]. The proposed Dengue Health Intelligence component therefore supports context-aware processing of dengue-related information rather than relying solely on keyword matching. It integrates multilingual question answering, information classification, evidence retrieval, claim verification, misinformation analysis, and safety-oriented severity assessment. The system supports **Sinhala, Singlish, and English** interactions and is designed as a decision-support tool rather than a replacement for professional medical assessment.

E. Research Gap and Integration of AI-Based Dengue Decision Support

The reviewed literature highlights the potential of Machine Learning for outbreak prediction, XAI for improving transparency, Reinforcement Learning for decision support, NLP for health-information processing, and Computer Vision for image-based analysis. However, these technologies are often studied separately, creating a gap in integrating **predictive, explanatory, prescriptive, communicative, and field-level AI capabilities** within a unified dengue public-health platform. The proposed research addresses this gap by integrating dengue outbreak prediction, SHAP-based XAI, Q-Learning-based intervention recommendation, multilingual Dengue Health Intelligence, and AI-assisted PHI field inspection into a

single mobile decision-support system. This integration supports a complete workflow from **risk prediction and explanation to intervention recommendation, health-information delivery, and field-level verification**.

III. METHODOLOGY

The proposed research presents an integrated Artificial Intelligence (AI)-based mobile decision-support system for dengue outbreak management. It combines Machine Learning (ML), Explainable Artificial Intelligence (XAI), Reinforcement Learning (RL), Natural Language Processing (NLP), evidence-based information retrieval, and Computer Vision to support dengue prevention, surveillance, and public-health decision-making. The system consists of four major components: dengue outbreak prediction, XAI-based prediction explanation and data visualization, RL-based intervention recommendation, and Dengue Health Intelligence with AI-assisted Public Health Inspector (PHI) field inspection. These components are integrated through backend services and delivered through a unified Flutter-based mobile application. The overall methodology was designed to transform dengue surveillance data, environmental information, user-submitted health queries, and field inspection images into meaningful decision-support outputs. Historical and environmental data are used to predict dengue outbreak risk, while XAI techniques explain the factors influencing the prediction. Based on the identified outbreak and environmental conditions, the RL component recommends appropriate interventions. The Dengue Health Intelligence component provides multilingual, evidence-based analysis of dengue-related questions and claims, while the PHI field inspection module supports AI-assisted detection and verification of potential mosquito breeding-risk conditions.

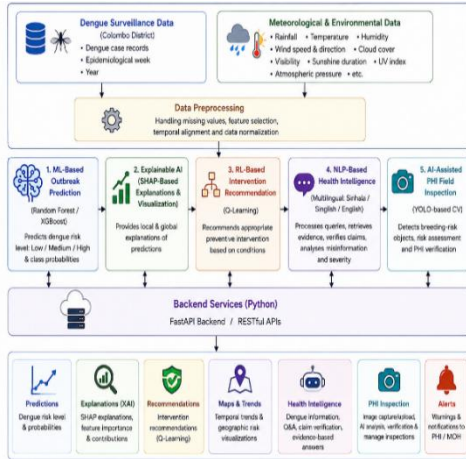


Fig. 1. Overall system architecture

A. Overall System Architecture

As illustrated in **Fig. 1**, the proposed system follows an integrated architecture that connects multiple AI-based components through backend services and a Flutter-based mobile application. The workflow begins with the collection and preprocessing of historical dengue surveillance, meteorological, and temporal data. The processed data are

supplied to the ML model, which classifies dengue outbreak conditions into **Low, Medium, or High-risk** levels. The prediction and associated probabilities are then provided to the XAI and intervention recommendation components. The XAI component applies SHAP-based techniques to identify the environmental and temporal factors influencing each prediction, providing both global feature importance and local explanations for individual predictions.

The RL component receives relevant outbreak and environmental conditions and uses a learned decision policy to recommend appropriate dengue prevention or control action. The Dengue Health Intelligence component processes user-submitted questions, statements, and health-related claims in Sinhala, Singlish, and English. The backend determines the appropriate processing route and applies mechanisms such as question answering, evidence retrieval, claim verification, misinformation analysis, current-information retrieval, and severity assessment. The PHI Field Inspection component enables field officers to capture or upload inspection photographs together with relevant field information and geographical coordinates. The submitted image is analyzed through an AI-based Vision service to identify potential breeding-risk objects or environmental conditions. The generated result is subsequently reviewed by the PHI officer before public-health actions are initiated. All components communicate with the mobile application through RESTful APIs, allowing users and relevant public-health personnel to access the system through a unified interface.

B. Data Collection: Source Method

The primary data sources comprised historical dengue surveillance records and meteorological observations for the Colombo District. Dengue data were obtained from the **National Dengue Control Unit (NDCU), Ministry of Health, Sri Lanka**, while meteorological data were obtained from the **Department of Meteorology, Sri Lanka** [12], [13]. The environmental variables included rainfall, humidity, maximum, minimum, and mean temperature. These datasets were integrated to capture the relationship between dengue occurrence and environmental conditions, consistent with previous dengue prediction studies [2], [3]. **Table I** summarizes the primary data sources used in the study.

Table 1: Primary Data Sources

Data Category	Source	Main Variables
Dengue surveillance data	National Dengue Control Unit (NDCU)	Dengue cases, year, epidemiological week
Meteorological data	Department of Meteorology, Sri Lanka	Rainfall, humidity, maximum temperature, minimum temperature, mean temperature
Temporal variables	Derived during preprocessing	Year, week, month

C. Data Collection Methods

The collected dengue surveillance and meteorological datasets were temporally aligned using **year** and **epidemiological week** to associate epidemiological and environmental observations from the same period. The data were cleaned by addressing duplicate records, missing values, inconsistencies, incompatible data types, and numerical formatting issues. Missing values were imputed to preserve the chronological continuity required for time-based prediction. Relevant temporal and meteorological features, including lagged variables representing previous dengue and environmental conditions, were then prepared where applicable. The dataset was organized chronologically, and dengue risk was categorized into **Low, Medium, and High** levels based on case counts. To reduce target leakage and support realistic future-risk prediction, the current dengue case count was excluded from the predictive features.

D. Data Analysis Methods and Techniques

The **Machine Learning component** used Random Forest and XGBoost to predict dengue outbreak risk from historical and meteorological variables [4], [5]. A chronological train-test approach was applied, and performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. The **XAI component** used SHAP to provide local and global explanations of model predictions [6]. The **Reinforcement Learning component** used Q-Learning to recommend dengue prevention and control actions based on identified risk conditions [7], [15]. The **NLP component** used DistilBERT to process and classify English and Sinhala dengue-related information, including symptoms, prevention, treatment advice, guidelines, outbreak warnings, and misinformation [8], [9].

E. Tools and Materials Used

Python was used for data preprocessing and AI model development. Scikit-learn supported the Machine Learning implementation [18], while SHAP was used for explainability. The NLP component was developed using DistilBERT and transformer-based libraries. FastAPI was used to develop backend RESTful API services, and Flutter was used to implement the mobile application. The main technologies used in the study are presented in **Table 2**.

Table 2: Tools and Technology

Category	Tools
Programming	Python, Dart
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Explainable AI	SHAP
NLP	DistilBERT
Backend	FastAPI
Mobile Application	Flutter

F. Bias and Mitigation Strategies

Potential issues such as class imbalance, missing data, and temporal data leakage were considered during the study. Class-specific evaluation metrics, including precision, recall, and F1-score, were used in addition to overall accuracy. A chronological train-test split was applied to reduce the risk of using future observations during training. Missing values and inconsistent records were addressed during preprocessing. Official data sources and verified health-information sources were used to improve data reliability. The developed system is intended to support public health decision-making rather than replace professional judgment.

G. Visualization of Results

The results were presented through a Flutter-based mobile application. The prediction component displayed dengue risk levels and associated probabilities, while the XAI component presented the factors influencing each prediction. The application also included temporal trend visualizations, geographic risk information, intervention recommendations, chatbot services, and alert notifications. The visualization layer transforms the output of the AI-based components into understandable information for users and supports the overall dengue decision-support process.

IV. RESULTS AND DISCUSSION

The proposed system integrates Machine Learning-based dengue outbreak prediction, Explainable Artificial Intelligence, Reinforcement Learning-based intervention recommendation, and NLP-based health intelligence into a single mobile decision-support platform. **This integrated approach combines prediction, explanation, intervention recommendation, and health-information support to provide a more comprehensive and proactive solution for dengue surveillance and control.**

A. Machine Learning-Based Dengue Outbreak Prediction

The Machine Learning component was evaluated using a chronological dataset, with 2015–2024 observations used for model training and 2025 observations retained as an independent test set. Random Forest and XGBoost were evaluated for classifying dengue outbreak risk into High, Low, and Medium categories (See **Table 3**).

Table 3: Model Performance comparison

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	86.35%	86.37%	86.35%	86.36%
XGBoost	85.58%	85.65%	85.58%	85.60%

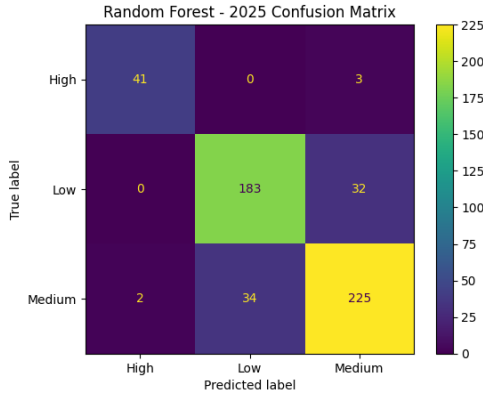


Fig. 2. Confusion matrix of the Random Forest model

B. Future Dengue Risk Estimation

The selected Random Forest model was applied to **2026 Weeks 1–22**, covering 220 area-week observations across the Colombo District.

Table 4: 2026 Predicted Risk Distribution

Risk Level	Records	Percentage
Medium	143	65.00%
Low	54	24.55%
High	23	10.45%
Total	220	100%

Medium risk was the dominant predicted category, representing 65.00% of the evaluated observations, while 10.45% were classified as High risk.

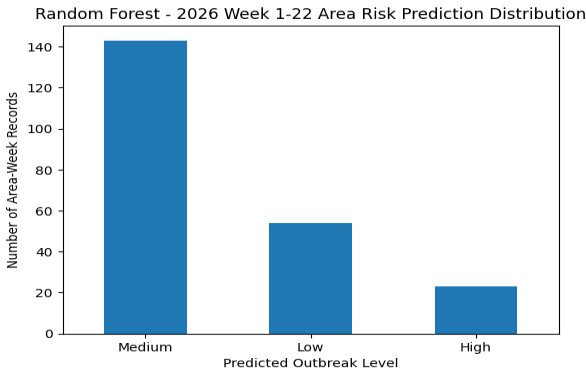


Fig. 3. Predicted dengue-risk distribution for 2026 Weeks 1–22

C. SHAP-Based Explainable Artificial Intelligence

SHAP was integrated with the selected Random Forest model to provide **global and local explanations** of dengue-risk predictions. Global SHAP analysis identifies the features with the greatest overall influence on model predictions, while local explanations show the contribution of individual features to a specific prediction. This improves the transparency and interpretability of the predicted risk levels.

D. Reinforcement Learning-Based Intervention Recommendation

The Q-Learning component provides risk-based intervention recommendations, including **monitoring, fumigation, public-health alerts, hospital preparation, and community clean-up activities**. This extends the system from risk prediction towards proactive intervention support by associating identified risk conditions with appropriate actions.

E. NLP-Based Dengue Health Intelligence

The NLP component uses **DistilBERT** to support dengue-related health information through the chatbot in English and Sinhala. The system provides information related to symptoms, prevention, treatment advice, official guidelines, outbreak warnings, and misinformation.

V. CONCLUSION

This research presents an integrated AI-based dengue outbreak prediction and intervention optimization system for the Colombo District, Sri Lanka. The proposed system combines Machine Learning, Explainable Artificial Intelligence, Reinforcement Learning, and Natural Language Processing within a unified mobile decision-support platform. Machine Learning predicts dengue risk as Low, Medium, or High using historical surveillance and meteorological data, while SHAP improves model transparency by explaining the factors influencing predictions. Q-Learning provides context-dependent intervention recommendations, and DistilBERT supports dengue-related information and misinformation classification. Through the Flutter mobile application, users can access predictions, XAI explanations, intervention recommendations, trends, geographic risk information, chatbot services, and alerts. Overall, the system demonstrates the feasibility of integrating predictive, explanatory, prescriptive, and communicative AI capabilities to support more proactive and data-driven dengue surveillance and control in Sri Lanka.

ACKNOWLEDGMENT

The research team acknowledges the support and guidance provided by the research supervisors and the contribution of the National Dengue Control Unit (NDCU) and the Department of Meteorology, Sri Lanka, to this research.

REFERENCES

- [1] World Health Organization, “Dengue and Severe Dengue,” World Health Organization. Accessed: Aug. 29, 2026. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue>
- [2] W. Hoyos, J. Aguilar, and M. Toro, “Dengue Models Based on Machine Learning Techniques: A Systematic Literature Review,” *Artificial*

- Intelligence in Medicine*, vol. 119, p. 102157, Sep. 2021, doi: 10.1016/j.artmed.2021.102157.
- [3] “Neural Networks for Dengue Prediction: A Systematic Review,” Google.com. Accessed: Aug. 29, 2026. [Online]. Available: https://scholar.google.com/citations?view_op=view_citation&hl=en&user=dSmNi-AAAAAJ&citation_for_view=dSmNi-AAAAAJ:2osOgNQ5qMEC
- [4] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, Oct. 2001.
- [5] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16*, vol. 1, no. 1, pp. 785–794, Aug. 2016, doi: 10.1145/2939672.2939785.
- [6] S. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” arXiv.org. Accessed: Aug. 29, 2026. [Online]. Available: <https://arxiv.org/abs/1705.07874v2>
- [7] C. J. C. H. Watkins and P. Dayan, “Q-Learning,” *Machine Learning*, vol. 8, no. 3–4, pp. 279–292, May 1992, doi: 10.1007/bf00992698.
- [8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding,” *Proceedings of the 2019 Conference of the North*, vol. 1, pp. 4171–4186, 2019, doi: 10.18653/v1/n19-1423.
- [9] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter,” arXiv.org. Accessed: Aug. 29, 2026. [Online]. Available: <https://arxiv.org/abs/1910.01108>
- [10] World Health Organization, “Global Vector Control Response 2017–2030,” www.who.int. Accessed: Aug. 29, 2026. [Online]. Available: <https://www.who.int/publications/i/item/9789241512978>
- [11] WORLD HEALTH ORGANIZATION, “Dengue Guidelines, for Diagnosis, Treatment, Prevention and Control,” www.who.int. Accessed: Aug. 29, 2026. [Online]. Available: <https://www.who.int/publications/i/item/9789241547871>
- [12] “Welcome to Epidemiology Unit Official Website,” www.epid.gov.lk. Accessed: Aug. 29, 2026. [Online]. Available: <https://www.epid.gov.lk/>
- [13] “National Dengue Control Unit,” www.dengue.health.gov.lk. Accessed: Aug. 29, 2026. [Online]. Available: <https://www.dengue.health.gov.lk/>
- [14] CDC, “Dengue,” Centers for Disease Control and Prevention. Accessed: Aug. 29, 2026. [Online]. Available: <https://www.cdc.gov/Dengue/>
- [15] H. Sutton, “Peter Morgan Sutton,” *BMJ*, vol. 348, no. mar31 11, pp. g2466–g2466, Mar. 2014, doi: 10.1136/bmj.g2466.
- [16] A. Vaswani *et al.*, “Attention Is All You Need,” Jun. 2017. Accessed: Aug. 29, 2026. [Online]. Available: <https://arxiv.org/pdf/1706.03762>
- [17] T. G. Dietterich, “Ensemble Methods in Machine Learning,” *Multiple Classifier Systems*, vol. 1857, pp. 1–15, 2000, doi: 10.1007/3-540-45014-9_1.
- [18] “(PDF) Scikit-Learn: Machine Learning in Python,” *ResearchGate*, Accessed: Aug. 29, 2026. [Online]. Available: https://www.researchgate.net/publication/51969319_Scikit-learn_Machine_Learning_in_Python
- [19] “TensorFlow: A System for Large-Scale Machine Learning | Request PDF,” *ResearchGate*. Accessed: Aug. 29, 2026. [Online]. Available: https://www.researchgate.net/publication/303657108_TensorFlow_A_system_for_large-scale_machine_learning
- [20] A. Paszke *et al.*, “PyTorch: An Imperative Style, High-Performance Deep Learning Library,” *arXiv (Cornell University)*, Dec. 2019, doi: 10.48550/arxiv.1912.01703.